Baseline Comparison and $\Delta P^{0.5}$ Stage-3 MLP

Hiroyasu–Arai, Naber–Siebers, Sparse GP, and five-seed external clean evaluation

Linan Jiang

Aalto University

9 May 2026

Frozen slide data snapshot: Thesis/generated/baseline_comparison_20260509/

Evidence Chain for This Deck

Question	Method	How to read it
Can a physics formula explain the data?	Hiroyasu–Arai and Naber–Siebers	Deterministic physics correlations; strong but rigid references.
Is a standard ML baseline already enough?	Sparse heteroscedastic GP	Non-parametric and uncertainty-aware; checks whether the MLP is just benefiting from small-data luck.
Is the neural network stable?	$\Delta P^{0.5}$ Stage-3, 5 seeds	Looks at seed-to-seed variation instead of a single-run headline.
Does it generalize to unseen nozzles?	Leave-one-nozzle-out	Removes an entire nozzle family from training to test true OOD interpolation/extrapolation pressure.
Is the uncertainty	Coverage, reliability, CRPS,	Separates ”small error” from

Metric Meaning: Accuracy, Tail, and Calibration

Point-prediction error

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (\hat{y}_i - y_i)^2}, \quad \mathrm{MAE} = \frac{1}{n}\sum_i |\hat{y}_i - y_i|.$$

RMSE is more sensitive to large errors; MAE is closer to the everyday meaning of average millimetre error.

$$\mathrm{Bias} = \frac{1}{n}\sum_i (\hat{y}_i - y_i)$$

is used to detect systematic over- or under-prediction.

Tail error and uncertainty

$$\mathrm{P95} = \mathrm{quantile}_{0.95}(|\hat{y} - y|)$$

means that 95% of the pointwise errors fall below this value, which directly reflects worst-case tail behaviour.

$$\mathrm{Coverage}_{k\sigma} = \Pr(|y - \mu| \le k\sigma).$$

If Gaussian predictions are well calibrated, $1\sigma/2\sigma$ should be close to 0.683/0.954. Values above nominal indicate over-coverage, which is usually conservative but less sharp.

Protocol Meaning: In-Pipeline vs External Clean

In-pipeline 5-seed bootstrap

Each seed re-runs the full Stage-1/2/3 pipeline, and the winner metrics are computed on that seed's post-evaluation test subset. The bootstrap object is the seed-level metric: the five seeds are resampled with replacement to produce an empirical distribution of the mean and a 95% interval. It answers: **does training randomness change the conclusion?**

External clean protocol

The trained models are then evaluated on one shared full-clean diagnostic set: $795\,561$ points, $33\,845$ trajectories. It answers: **which method is more accurate on the same large external clean set?**

Strict wording

These two protocols cannot be merged into a single number without care. This deck uses the in-pipeline result for seed stability and the external clean result for the headline model comparison.

One-Sentence Conclusion

Under the same external clean evaluation protocol

The five-seed mean of the Stage-3 MLP with $\Delta P^{0.5}$ feature engineering reaches RMSE $6.211\,\mathrm{mm} \pm 0.283\,\mathrm{mm}$ on $795\,561$ points and $33\,845$ trajectories.

Against physical baselines

- vs Hiroyasu–Arai: RMSE -37.7%, MAE -40.3%.
- vs Naber–Siebers: RMSE -29.7%, MAE -32.3%.
- vs sparse GP: RMSE -10.0%, MAE -11.9%.

Against the older MLP

- Old $\Delta P^{0.25}$, seed 42: RMSE $6.696\,\mathrm{mm}$.
- New $\Delta P^{0.5}$, five-seed mean: RMSE $6.211\,\mathrm{mm}$.
- Sparse GP mean: RMSE $6.898\,\mathrm{mm}$, MAE $5.206\,\mathrm{mm}$.

Source: snapshot/analysis_summary.md

Headline Comparison Table

Model	RMSE	MAE	Bias	P95	1σ cov.	2σ cov.
Hiroyasu–Arai calibrated	9.974	7.686	+0.625	20.493	0.536	0.886
Naber–Siebers delay	8.840	6.780	-0.344	17.922	0.620	0.893
Stage-3 MLP $\Delta P^{0.25}$, seed 42	6.696	4.875	-1.149	14.118	0.745	0.979
Stage-3 MLP $\Delta P^{0.5}$, 5-seed mean	6.211	4.587	-0.838	13.029	0.774	0.981
Sparse heteroscedastic GP, 5-seed mean	6.898	5.206	-1.184	14.365	0.670	0.950

How to position this in the thesis

The older MLP result should no longer be treated as the neural-network headline. It is now only a reference row before the ΔP exponent was changed from 0.25 to 0.5. The GP is a strong ML baseline: its calibration is close to nominal, but its accuracy is still below the new MLP five-seed mean.

Source: snapshot/headline_comparison.csv

Physical Baseline I: Hiroyasu–Arai / Zhou

Theory role

Hiroyasu–Arai writes spray penetration as a staged empirical law: the early phase is controlled by injection velocity, and after breakup it transitions to an entrainment-dominated square-root time scaling. The Zhou-style post-injection extension adds a $(t - t_i)^{1/4}$ branch after EOI.

$$S(t) = \begin{cases} k_v \sqrt{2\Delta P/\rho_l}\, t, & t < t_b, \\ k_p (\Delta P/\rho_a)^{1/4} d_n^{1/2} t^{1/2}, & t \geq t_b. \end{cases}$$

$$t_b = k_{bt} \frac{\rho_l d_n}{\sqrt{\rho_a \Delta P}}, \qquad S_{\mathrm{post}} \propto (\Delta P/\rho_a)^{1/4} d_n^{1/2} t_i^{1/4} (t - t_i)^{1/4}.$$

This study uses the calibrated version as the strong physics baseline; its uncertainty comes from residual statistics, not from an input-dependent learned $\sigma(x, t)$.

Physical Baseline II: Naber–Siebers

Theory role

Naber–Siebers is a mixing-limited penetration correlation. It folds the main operating-condition dependence into the physical scale A, then multiplies by a $\sqrt{t}$-type time growth.

$$S(t) = K \underbrace{\left(\frac{\Delta P}{\rho_a}\right)^{1/4} d_n^{1/2}}_{A_{\mathrm{NS}}} \sqrt{\max(t - t_d, 0)}.$$

- K: global scale; t_d: hydraulic/observation delay.
- Optional cone-angle variant can use video-measured θ, but the headline comparison uses the delay variant, which does not require an oracle visual feature.
- The point is to give the MLP an interpretable, low-degree-of-freedom, physically rigid baseline.

GP Learning Target: Swapping the Prior for MLP Regression

First write the problem as the same supervised-learning task as the MLP

Each sample is

$$\mathsf{x}_i = [\mathsf{t},\ \mathrm{tilt},\ \mathrm{plumes},\ \mathsf{t}_{\mathrm{inj}},\ \mathsf{p}_{\mathrm{ctrl}}]_i, \qquad \mathsf{z}_i = \frac{\mathsf{S}_i}{\mathsf{A}_i}.$$

where

$$\mathsf{A}_i = (\Delta \mathsf{P}_i)^{0.5} \rho_{\mathsf{a},i}^{-0.25} \mathsf{d}_{\mathsf{n},i}^{0.5}.$$

The MLP learns $\mathsf{x} \mapsto \mathsf{z}$. The GP baseline learns the same target and is then multiplied by A_i to return to millimetres.

The difference is only the function assumption

MLP: represent the function explicitly with parameters, $\mathsf{z}_\theta(\mathsf{x})$.
GP: do not write a fixed network first; instead place a probabilistic prior directly over the function $\mathsf{f}(\mathsf{x})$:

$$\mathsf{z}_i = \mathsf{f}(\mathsf{x}_i) + \epsilon_i.$$

GP Definition: Not a Curve, but a Family of Joint Gaussians

Finite-dimensional definition

If, for any finite input set $X = [x_1, \ldots, x_n]$, the function values satisfy

$$\mathbf{f}_X = [f(x_1), \ldots, f(x_n)]^\top \sim \mathcal{N}(\mathbf{m}_X, K_{XX}),$$

then

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')).$$

Mean function

$$(\mathbf{m}_X)_i = m(x_i)$$

This implementation uses a constant mean so that the data learn the global offset.

Kernel / covariance

$$(K_{XX})_{ij} = k(x_i, x_j)$$

The kernel determines how similar the function values at two input points should be.

Kernel Used Here: ARD Matérn-$5/2$

Distance is scaled by learned length-scales

$$r^2(x, x') = \sum_{j=1}^{5} \left(\frac{x_j - x'_j}{\ell_j} \right)^2 .$$

ℓ_j is the length-scale for feature dimension j. A large ℓ_j means the function is insensitive to that feature; a small ℓ_j means even small changes in that feature can move the prediction.

Matérn-$5/2$ covariance

$$k_\theta(x, x') = \sigma_f^2 \left(1 + \sqrt{5} r + \frac{5r^2}{3} \right) \exp(-\sqrt{5} r).$$

It is smoother than a rough kernel, but less overly smooth than an RBF kernel, which suits spray-penetration functions with transient behaviour.

Code mapping: `ScaleKernel(MaternKernel(nu=2.5, ard_num_dims=5))`.

Exact GP Derivation: Joint Gaussian over Train and Test Points

Homoscedastic observation noise, starting from the simplest case

$$\mathbf{z} = \mathbf{f}_{\mathsf{X}} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_{\mathsf{n}}^2 \mathsf{I}).$$

For test inputs X_*, the joint distribution is

$$\begin{bmatrix} \mathbf{z} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mathbf{m}_{\mathsf{X}} \\ \mathbf{m}_* \end{bmatrix}, \begin{bmatrix} \mathsf{K}_{\mathsf{XX}} + \sigma_{\mathsf{n}}^2 \mathsf{I} & \mathsf{K}_{\mathsf{X}*} \\ \mathsf{K}_{*\mathsf{X}} & \mathsf{K}_{**} \end{bmatrix} \right).$$

The conditional Gaussian formula is the GP prediction formula

$$\mu_* = \mathbf{m}_* + \mathsf{K}_{*\mathsf{X}}(\mathsf{K}_{\mathsf{XX}} + \sigma_{\mathsf{n}}^2 \mathsf{I})^{-1}(\mathbf{z} - \mathbf{m}_{\mathsf{X}}),$$
$$\Sigma_* = \mathsf{K}_{**} - \mathsf{K}_{*\mathsf{X}}(\mathsf{K}_{\mathsf{XX}} + \sigma_{\mathsf{n}}^2 \mathsf{I})^{-1}\mathsf{K}_{\mathsf{X}*}.$$

These two lines are just ”weighted averaging from similar training points, plus uncertainty.”

Why Not Use an Exact GP Directly on This Dataset?

Hyperparameter learning comes from the marginal likelihood

Let $\mathsf{K}_\mathsf{y} = \mathsf{K}_{\mathsf{XX}} + \sigma_\mathsf{n}^2 \mathsf{I}$. Then

$$\log \mathsf{p}(\mathbf{z}|\mathsf{X}, \theta) = -\frac{1}{2}(\mathbf{z} - \mathbf{m})^\top \mathsf{K}_\mathsf{y}^{-1}(\mathbf{z} - \mathbf{m}) - \frac{1}{2}\log|\mathsf{K}_\mathsf{y}| - \frac{\mathsf{n}}{2}\log 2\pi.$$

The first term rewards fit, the second penalizes an overly complex covariance, and the third is constant.

The bottleneck

To compute $\mathsf{K}_\mathsf{y}^{-1}\mathbf{z}$ and $\log|\mathsf{K}_\mathsf{y}|$, one usually needs a Cholesky factorization:

$$\mathsf{K}_\mathsf{y} = \mathsf{L}\mathsf{L}^\top, \qquad \text{cost} = \mathsf{O}(\mathsf{n}^3), \quad \text{memory} = \mathsf{O}(\mathsf{n}^2).$$

Here each trajectory expands into many time points, and the number of training points per seed can reach the million range. So an exact GP is infeasible and we need a sparse variational GP.

Sparse GP: Compressing Function Information with Inducing Points

Core idea

Choose $m \ll n$ inducing inputs

$$Z = [z_1^{(u)}, \dots, z_m^{(u)}], \qquad \mathbf{u} = f(Z).$$

They are not training labels, but support points of the function. Training learns the function distribution around these points and then uses them to explain all n training samples.

Conditional structure

The GP prior gives

$$p(\mathbf{f}, \mathbf{u}) = p(\mathbf{f}|\mathbf{u})p(\mathbf{u}), \qquad p(\mathbf{u}) = \mathcal{N}(\mathbf{m}_Z, K_{ZZ}).$$

If $m = 256$, the main matrix operations shift from $n \times n$ to $m \times m$ and minibatch covariance terms.

This implementation initializes 256 inducing points with K-means over the training inputs and allows their positions to keep learning.

Variational GP: Replacing the Posterior $p(u|D)$ with Optimizable $q(u)$

Variational distribution

The exact posterior $p(\mathbf{u}|\mathcal{D})$ is hard to compute directly, so we set

$$q(\mathbf{u}) = \mathcal{N}(\mathbf{m}_u, S_u),$$

where $\mathbf{m}_u$ and S_u are trainable parameters. For any training-point function values:

$$q(\mathbf{f}) = \int p(\mathbf{f}|\mathbf{u})q(\mathbf{u})\,d\mathbf{u}.$$

ELBO: optimize a computable lower bound

$$\mathcal{L} = \sum_{i=1}^{n} \mathbb{E}_{q(f_i)}[\log p(z_i|f_i)] - \mathrm{KL}(q(\mathbf{u}) \,\|\, p(\mathbf{u}))\,.$$

The first term demands that the predictions explain the data; the second prevents $q(u)$ from drifting too far from the GP prior. In minibatch training, the batch likelihood term is scaled by $n/|\mathcal{B}|$.

The Heteroscedastic GP in This Thesis: Two SVGPs in Series

Stage 1: mean SVGP

$$f_\mu(x) \sim \mathcal{GP}(m_\mu, k_\mu), \qquad z_i = f_\mu(x_i) + \epsilon_i.$$

After training we get

$$\hat{\mu}_z(x) = \mathbb{E}[f_\mu(x)].$$

Stage 2: log-variance SVGP

First use the mean GP to compute residuals on the training set:

$$r_i = z_i - \hat{\mu}_z(x_i).$$

Then build a variance target:

$$g_i = \log(r_i^2 + \varepsilon).$$

The second SVGP learns

$$f_v(x) \approx g, \qquad \hat{\sigma}_z^2(x) = \exp(\hat{f}_v(x) + b).$$

b is a log-variance bias correction fitted on validation data.

From Scaled GP Output Back to Millimetre Penetration

Scaled space

GP training and prediction both happen in

$$z = \frac{S}{A_{\mathrm{scale}}}$$

space, so the outputs are

$$\hat{\mu}_z(x), \qquad \hat{\sigma}_z(x).$$

Physical space

Back in penetration units:

$$\hat{\mu}_S(x) = A_{\mathrm{scale}}(x)\hat{\mu}_z(x), \qquad \hat{\sigma}_S(x) = A_{\mathrm{scale}}(x)\hat{\sigma}_z(x).$$

RMSE, MAE, P95, and coverage are all computed using $\hat{\mu}_S, \hat{\sigma}_S$ and the true S in millimetres.

One implementation detail

The headline GP uses residual log-variance as the main uncertainty source; `include_mean_posterior_var=false`, meaning we do not add the mean GP posterior variance

Mapping the Formulas to Project Code

Mathematical object	Implementation	Interpretation
x	`a_only` features	time, tilt, plume count, injection duration, control backpressure.
$z = S/A$	`target_scaled`	The same $\Delta P^{0.5}$ scale used by the Stage-3 MLP.
k_θ	ARD Matérn-$5/2$	One length-scale per feature dimension.
$Z, \mathbf{u}$	256 inducing points	K-means initialization, with trainable positions.
$q(u)$	Cholesky variational distribution	GPyTorch sparse variational posterior.
$\mathcal{L}$	`VariationalELBO`	Minibatch Adam, mean and logvar each trained for up to 80 epochs.
$\sigma(x)$	Second SVGP on $\log r^2$	Two-stage heteroscedastic uncertainty.

Code entry point: `MLP/MLP_training/run_gp_baseline.py`

Why Is This GP Baseline Fair, Yet Still Worse than the MLP?

Fairness

The GP uses the same split, the same five seeds, the same A_{scale} target, the same controllable input features, and the same external clean diagnostic set as the MLP.

GP strength

The kernel prior imposes a strong smoothness assumption, the SVGP gives probabilistic prediction, and the two-stage log-variance model lets uncertainty vary with the input. So this is not a weak baseline; it does bring coverage close to nominal.

MLP strength

The Stage-3 MLP is more than a plain regressor: it inherits the full trajectory structure of the fitted target and then refines the raw CDF in the early reliable region. The GP baseline learns the same pointwise targets but does not have that teacher/raw two-stage training mechanism.

Can the GP Beat the MLP?

Accuracy: no

Metric	MLP	GP
RMSE	6.211	6.898
MAE	4.587	5.206
P95	13.029	14.365

Relative to the new MLP, the GP is worse by 11.1% / 13.5% / 10.2% in RMSE / MAE / P95.

Calibration: the GP is closer to nominal

Metric	MLP	GP
1σ cov.	0.774	0.670
2σ cov.	0.981	0.950

Gaussian nominal is 0.683/0.954. The GP uncertainty is closer to nominal, while the MLP is conservative on the clean set.

Conclusion

The GP is a strong baseline, especially because it shows the uncertainty comparison is fair. But under the same external clean protocol, it does not beat the $\Delta P^{0.5}$ Stage-3 MLP.

Source: snapshot/headline_comparison.csv

OOD / LONO: What the Protocol Means

Leave-one-nozzle-out protocol

LONO removes all samples from one nozzle family / experiment label from training and tests only on that held-out group. It is stricter than an ordinary group-aware split because the test nozzle is completely unseen during training.

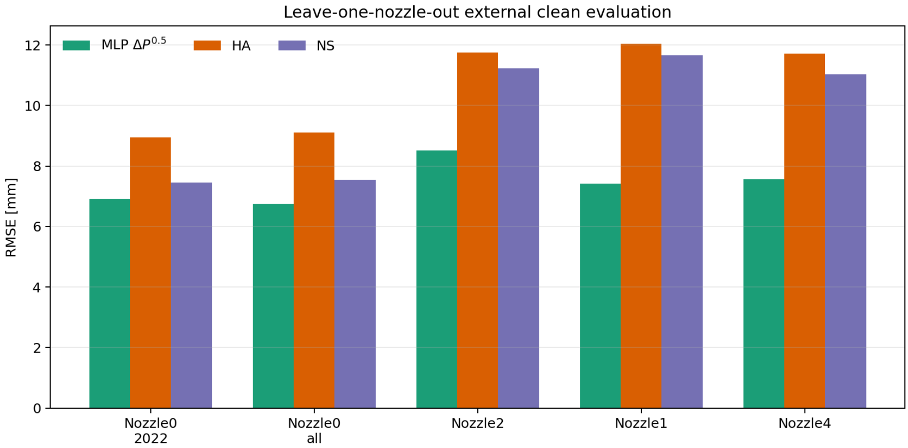
What it answers

External clean eval mainly checks in-distribution performance; LONO checks whether the model can preserve physical scaling and residual learning on unseen nozzle families. The statistical object here is fold-to-fold variation, not seed-to-seed variation.

Current 5-fold result

MLP $\Delta P^{0.5}$: RMSE 7.435 ± 0.692 mm; HA: 10.713 ± 1.542 mm; NS: 9.784 ± 2.102 mm. Relative to HA/NS, RMSE drops by about 30.6% / 24.0%.

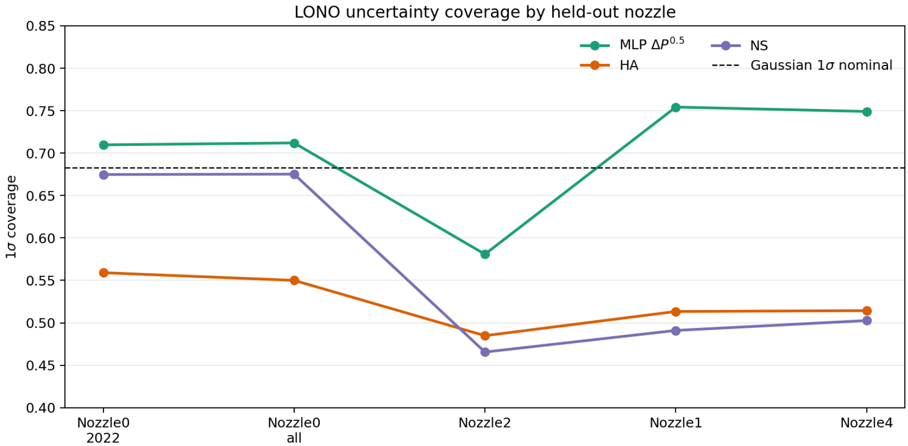
LONO Result: RMSE for Each Held-Out Nozzle



Across the five held-out folds, the MLP is below both HA and NS. The hardest fold is BC20241017_HZ_Nozzle2, where the MLP RMSE is 8.517 mm, still below HA 11.754 mm and NS 11.236 mm.

Source: MLP/runs_mlp/lono_pipeline_20260509_215613/per_fold.csv

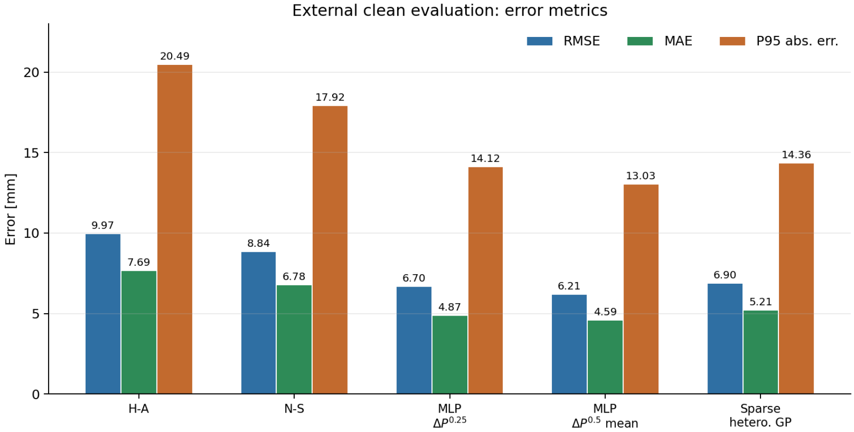
LONO Result: Does Coverage Collapse under OOD?



The MLP's mean 1σ coverage is 0.701, close to the Gaussian nominal 0.683. The Nozzle2 fold is clearly under-covered, which means that nozzle family remains an OOD risk point instead of being hidden by the average.

Source: MLP/runs_mlp/lono_pipeline_20260509_215613/aggregate.csv

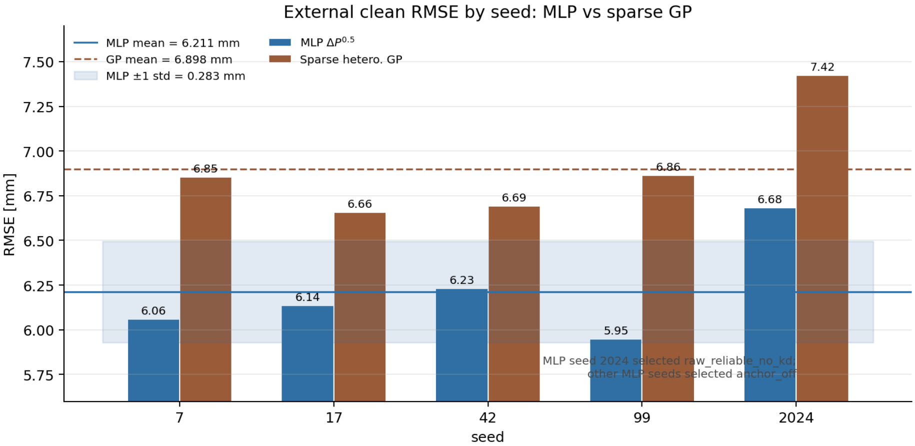
Error Metrics: Physical Baseline, GP, and MLP



Under this clean evaluation protocol, the sparse GP is clearly better than the physical baselines, but it still does not surpass the $\Delta P^{0.5}$ Stage-3 MLP in RMSE, MAE, or tail P95 error.

Source: snapshot/headline_comparison.csv

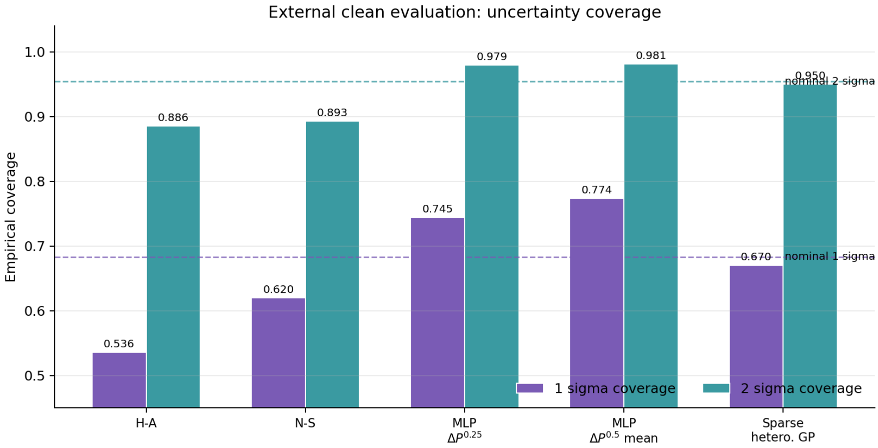
Five-Seed External Evaluation: MLP vs GP



The five-seed RMSE mean for the $\Delta P^{0.5}$ MLP is 6.211 mm; the GP five-seed external mean is 6.898 mm. Even the GP's best seed remains above the MLP five-seed mean.

Source: snapshot/new_mlp_5seed_aggregate.csv

Coverage: GP Is Closer to Nominal, MLP Is More Accurate



The new MLP has $1\sigma/2\sigma$ coverage of 0.774/0.981, which is over-coverage on the clean set; the GP is 0.670/0.950, closer to the Gaussian nominal 0.683/0.954. So the GP wins on calibration closeness, while the MLP wins on accuracy.

Source: snapshot/headline_comparison.csv

Recommended Thesis Wording

Abstract / introduction version

$\Delta P^{0.5}$ feature engineering reduced the Stage-3 MLP RMSE from 6.70 mm for the previous $\Delta P^{0.25}$ seed-42 model to 6.21 mm $\pm$ 0.28 mm over five full-clean external evaluations.

Results version

Under the same $795\,561$-point clean diagnostic protocol, the $\Delta P^{0.5}$ Stage-3 MLP reduced RMSE by 37.7% relative to the calibrated Hiroyasu–Arai correlation, by 29.7% relative to the Naber–Siebers delay model, and by 10.0% relative to a sparse heteroscedastic GP.

Source: snapshot/DATA_SOURCES.md

GP and Stability Caveat

GP baseline version

The sparse heteroscedastic GP reached near-nominal uncertainty coverage (0.670/0.950 for $1\sigma/2\sigma$), but its external-clean RMSE, MAE, and P95 error were 11.1%, 13.5%, and 10.2% higher than the $\Delta P^{0.5}$ Stage-3 MLP five-seed mean.

Limitations version

The five-seed aggregate is a $\Delta P^{0.5}$ Stage-3 pipeline aggregate, not a strict same-ablation seed study: seed 2024 selected `raw_reliable_no_kd`, whereas the other four seeds selected `anchor_off`.

Source: snapshot/DATA_SOURCES.md

Data Sources and Refresh Entry Point

Item	Path
Frozen snapshot	Thesis/generated/baseline_comparison_20260509/
Source report	MLP/baseline/comparison_reports/gp_vs_mlp_20260509/
Headline table	Thesis/generated/baseline_comparison_20260509/headline_comparison.csv
Five-seed aggregate	Thesis/generated/baseline_comparison_20260509/new_mlp_5seed_aggregate.csv
Per-seed eval rows	Thesis/generated/baseline_comparison_20260509/per_seed_new_mlp_eval.csv
GP bootstrap	Thesis/generated/baseline_comparison_20260509/gp_bootstrap_summary.json
LONO result	MLP/runs_mlp/lono_pipeline_20260509_215613/
Figure builder	Thesis/generated/baseline_comparison_20260509/make_figures.py

Refresh flow

Update the comparison-report CSVs → overwrite the CSVs in this snapshot → run .venv/Scripts/python.exeThesis/generated/baseline_comparison_20260509/make_figures.py → rebuild these slides.

Protocol label